

guide

Markdown

📈 Stock Price Prediction System (Live API & Auth)

Advanced Stock Price Prediction system built with **Deep Learning (LSTM)**, featuring live data fetching from Yahoo Finance, real-time news, and a secure user authentication system.

🚀 Features

- * **User Authentication:** Secure Login and Register system using `bcrypt` password hashing and `SQLite` database.
- * **Live Data Fetching:** Dynamic stock selection (20+ stocks) fetching real-time data via the `yfinance` API.
- * **Deep Learning Model:** Long Short-Term Memory (LSTM) network for time-series forecasting.
- * **Real-time News:** Fetches the latest market news for the selected ticker.
- * **Interactive Dashboard:** Responsive UI with real-time charts (Chart.js), performance metrics (MAE, RMSE, MAPE), and training epoch logs.

🛠️ Tech Stack

- * **Frontend:** HTML5, Tailwind CSS, JavaScript (ES6+), Chart.js.
- * **Backend:** FastAPI (Python), Uvicorn.
- * **ML/Data:** TensorFlow/Keras (LSTM), Scikit-Learn, Pandas, Numpy.
- * **Database:** SQLite.
- * **Finance API:** yfinance.

📦 Installation & Setup

1. Clone or Extract the Project

Ensure all files are in the same directory.

2. Setup Virtual Environment (Recommended)

```
```bash
```

```
python3 -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate
```

### 3. Install Dependencies

Bash

```
pip install fastapi uvicorn tensorflow pandas numpy scikit-learn yfinance
bcrypt
```

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## How to Run

### Step 1: Start the Backend (API)

Open a terminal and run:

Bash

```
python backend/main.py
```

*The server will start at `http://localhost:8000`*

### Step 2: Start the Frontend

Open a **new** terminal tab, navigate to the `frontend` folder, and run:

Bash

```
cd frontend
python3 -m http.server 5500
```

### Step 3: Access the App

Open your browser and navigate to: `http://localhost:5500/login.html`

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## Model Evaluation

The system evaluates the LSTM model using:

- **MAE (Mean Absolute Error):** Average magnitude of errors.
  - **RMSE (Root Mean Squared Error):** Penalizes larger errors.
  - **MAPE (Mean Absolute Percentage Error):** Accuracy in percentage terms.
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## Project Structure

- `backend/main.py` : FastAPI server & LSTM Logic.
- `frontend/` : Dashboard, Login, and Register pages.
- `users.db` : SQLite database for user accounts.
- `requirements.txt` : List of required Python libraries.